

SIH 2026 — Problem Statement 26170

AI-Driven Anomaly Detection in Component Burn-In & Screening

Comprehensive Research, Prior-Art, Solution Architecture & Dataset Report

Prepared as a technical reference for solution design, Claude Code implementation, and SIH presentation strategy.

Executive conclusion

Recommended direction: implement the complete PS requirement, but extend it into a reliability-intelligence pipeline rather than stopping at a generic anomaly detector. The target architecture is: absolute specification check → lot-adaptive baseline → trajectory/drift analysis → 168h prediction → uncertainty estimation → latent-defect risk fusion → PASS/REVIEW/REJECT → engineer-readable explanation.

This is a **recommended differentiation strategy, not a claim of proven academic novelty**. Existing literature already covers AI for semiconductor burn-in, lot-health assessment, degradation prognostics, and probabilistic prediction. The defensible opportunity is to investigate whether a PS-specific integration of these elements creates meaningful differentiation and measurable improvement.

1. What PS 26170 is actually asking

The PS describes a failure mode that static screening can miss: a component may remain below an absolute datasheet limit while showing abnormal drift relative to its lot and developing a trajectory associated with future failure. Module A asks for dynamic, population/lot-relative anomaly detection. Module B asks for early prediction of Value_168h from Value_0h and Value_24h and early rejection when predicted drift exceeds a safety slope. Explainability is also explicitly evaluated.

Engineering interpretation: the problem is not merely 'classify bad components'. It is 'identify components that are currently passing but whose behavior is inconsistent with their peers and/or indicates future degradation, while minimizing catastrophic false negatives.'

2. Two solution directions compared

Dimension	Approach A — Direct PS	Approach B — Reliability Intelligence
Core flow	Anomaly detection + 168h regression	PS modules + lot context + trajectory + uncertainty + risk fusion
PS compliance	High	High
Differentiation	Low–moderate	Higher, if validated
Likely methods	Z-score, Isolation Forest, XGBoost/LSTM	Robust statistics + degradation model + probabilistic prediction + decision layer
Latent-defect focus	Partial	Explicit
Explainability	Can be added	Built into decision evidence
False-negative handling	Threshold tuning	Risk-based escalation / REVIEW state
Main risk	Looks generic	Can become over-engineered
Recommendation	Baseline	Primary target architecture

Decision: Approach B should be the target architecture, while Approach A should remain the first baseline implementation.

3. Proposed architecture

Therefore the safest differentiation statement is: 'A proposed reliability-oriented framework that unifies lot-relative anomaly detection and early degradation prediction into an uncertainty-aware latent-defect risk decision.' This should be called a *proposed/differentiated framework* until a dedicated patent and literature search confirms that the exact formulation is not already disclosed.

6. Previous ISRO/SIH strategy — what we can learn

ISRO's official SIH 2024 problem list included SAR image colorization and highway/service-road map matching among other technical problems. Publicly documented examples show that successful solutions can add domain-specific technical structure rather than merely implement a generic model. For example, a documented SIH 2024 winning implementation for the map-matching problem reported combining road-network processing/OSRM with a Hidden Markov Model and trajectory analysis. The lesson is not that HMM is required; it is that the winning pattern can be: **literal PS requirement + domain knowledge + appropriate algorithmic combination + useful operational output.**

7. Prior-art findings that constrain our claims

Prior work	Relevant finding	Implication
ScienceDirect / Computers & Industrial Engineering 2025	Die Engineering 2025 factor burn-in quality prediction; PAA + PCA + DoD + PCA + Pearson + probabilistic SVM.	Bayesian optimization
ESREL 2023	Lot-specific health factor + binary classifier + LSTM autoencoder	Do not claim health assessment new.
NASA MOSFET prognostics	Run-to-failure thermal-overstress MOSFET data; Gaussian-process degradation model	Bayesian prognostics have been studied.
NASA IGBT accelerated aging	Accelerated-aging data from six IGBT devices with multiple electrical variables	Useful for validation source; not an SIH dataset.

8. Dataset strategy

The current public PS material located for this research does not provide an official dataset. Therefore development should use a two-track strategy: (1) real public aging/prognostics data for methodological development, and (2) a controlled PS-shaped synthetic dataset for testing the exact lot-relative/sparse-time-series problem.

Dataset	Use	Link
NASA MOSFET Thermal Overstress Aging	Primary degradation/prognostics development; run-to-failure power MOSFET experiments under thermal overstress	NASA PCOE repository and direct ZIP
NASA IGBT Accelerated Aging	Cross-device robustness experiments; accelerated thermal aging with NASA Open Data Portal	NASA Open Data Portal
Synthetic PS26170 dataset	Exact 0h/24h/96h/168h, lot-relative anomaly and latent-defect scenarios	Create internally; do not misrepresent as official

NASA MOSFET: <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/> | Direct ZIP: <https://phm-datasets.s3.amazonaws.com/NASA/13.+MOSFET+Thermal+Overstress+Aging.zip>

NASA IGBT: <https://data.nasa.gov/dataset/insulated-gate-bipolar-transistor-igbt-accelerated-aging>

ISRO SIH 2024 official problem reference: <https://vedas.sac.gov.in/en/sih2024.html>

9. Synthetic dataset design

Create a PS-shaped dataset with fields such as Component_ID, Lot_ID, Device_Family, Temperature, Value_0h, Value_24h, Value_96h, Value_168h, Spec_Min, Spec_Max and Failure_Label. Generate healthy trajectories, high-but-stable components, gradual latent degradation, strong degradation, sudden failure, and measurement anomalies. The most important synthetic cases are components that pass the specification but are abnormal relative to their lot, and components that have a high value but remain consistent with their lot trajectory.

10. Required model baselines

- Static specification limits
- Robust Z-score / MAD or comparable robust statistics
- Isolation Forest
- One-class / density-based baseline where appropriate
- Simple linear or parametric drift model
- Tree-based regression baseline such as XGBoost
- One probabilistic prediction baseline
- Full proposed reliability-intelligence architecture

11. Evaluation protocol

Anomaly detection: recall, precision, F1, PR-AUC, false-negative rate. **Prediction:** MAE, RMSE and appropriate error metrics. **Uncertainty:** prediction-interval coverage/calibration. **Operational:** defect escape rate, false rejection rate, review rate and percentage automatically cleared.

Because the PS explicitly treats false negatives as catastrophic, model selection should not be driven by accuracy alone. Thresholds and decision rules should reflect asymmetric consequences and preserve a human-review pathway.

12. Mandatory ablation study

- Static limits
- Static + lot statistics
- Static + lot + trajectory

→ Static + lot + trajectory + 168h prediction

→ Previous + uncertainty

→ Full latent-defect risk engine

The purpose is to demonstrate that each layer contributes measurable value. If an advanced layer does not improve performance or calibration, remove it rather than retaining it as buzzword-driven complexity.

13. What not to claim in the SIH presentation

✗ “First AI system for semiconductor burn-in.”

✗ “First latent-defect detection system.”

✗ “Novel use of uncertainty.”

✗ “Novel use of lot-level analysis.”

✗ “Novel use of LSTM/Transformer.”

✗ “Patent-pending” unless an actual filing exists.

Safer positioning: “A proposed reliability-oriented framework that moves from specification-based screening toward lot-relative, trajectory-based and uncertainty-aware latent-defect screening.”

14. Recommended SIH demo

Show a component that is below the official limit and therefore technically passes. Then reveal that its value is far outside the lot distribution, its early trajectory is abnormal, and its predicted 168h value is moving toward the safety boundary. Display the prediction interval and the evidence behind the risk decision. Finally show **REVIEW** rather than pretending the model has perfect certainty.

15. Claude Code implementation instructions

- Inspect the real dataset first; do not assume the schema.
- Build a data dictionary and identify actual temporal resolution.
- Check whether lot IDs, device IDs and failure labels exist.
- Prevent 168h leakage into the 0h/24h predictor.
- Split by component/lot appropriately to avoid leakage.
- Implement baselines before advanced models.
- Add each proposed layer incrementally.
- Track all experiments and metrics.
- Perform ablation and robustness tests.
- Do not add deep learning unless it improves the evidence.
- Do not fabricate failure labels or physics.
- Preserve an untouched final test set.

16. Final conclusion

Approach B is the recommended solution direction. It is more closely aligned with the latent-defect engineering problem and offers a stronger differentiation narrative than a generic anomaly detector. However, its value must be demonstrated empirically. The project should treat the individual methods as established building blocks and focus the research on a rigorous, PS-specific reliability decision framework.

Target positioning:

“From static pass/fail screening to uncertainty-aware, lot-relative, trajectory-based latent-defect risk screening.”

Research sources

- ISRO/SAC — Smart India Hackathon 2024 official ISRO problem statements: <https://vedas.sac.gov.in/en/sih2024.html>
- ScienceDirect — A data-driven modelling framework for predicting the quality of semiconductor devices to support burn-in decisions (Computers & Industrial Engineering, 2025): <https://www.sciencedirect.com/science/article/pii/S036083522500261X>
- ESREL 2023 — Process Data Analysis for Improved Burn-In Strategies Based on Complementary AI Models: <https://rpsonline.com.sg/proceedings/esrel2023/html/P620.html>
- NASA PCoE — Data Set Repository: <https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/>
- NASA Open Data — MOSFET prognostics: <https://data.nasa.gov/dataset/prognostics-of-power-mosfets-under-thermal-stress-accelerated-aging-using-data-driven-and->
- NASA Open Data — IGBT accelerated aging: <https://data.nasa.gov/dataset/insulated-gate-bipolar-transistor-igbt-accelerated-aging>
- PHM Society NASA mirror: <https://data.phmsociety.org/nasa/>

Research status: September 2026. Novelty claims remain subject to further literature/patent verification.